555 Timer Astable Multivibrator

Clock Pulse (Square Wave) Generator — ChronoBit'26

Team ConnectX

Overview

The digital clock's 1 Hz timing reference is generated using a 555 Timer IC configured as an astable multivibrator. In this mode, the 555 timer free-runs continuously without any external trigger, producing a repeating square wave at its output (pin 3). This square wave is used as the clock pulse that drives the seconds counter stage of the digital clock circuit.

Components Used

Component	Value / Type	Function
555 Timer IC	NE555	Generates the astable square-wave output
Resistor R1	68 kΩ	Charging resistor (sets HIGH time)
Resistor R2	10 kΩ	Charge/discharge resistor (sets LOW time)
Capacitor C	10 uF	Timing capacitor
LED	Standard output LED	Visual indicator of the output pulse

Theory of Operation

In astable mode, the 555 timer's internal comparators and flip-flop repeatedly charge and discharge the timing capacitor C through resistors R1 and R2, connected as shown below, with pins 2 (Trigger) and 6 (Threshold) tied together and connected to the capacitor.

- Charging phase (Output HIGH): The capacitor charges through R1 + R2 until its voltage reaches (2/3) VCC, at which point the internal comparator flips the output LOW and discharge begins.
- Discharging phase (Output LOW): The capacitor discharges through R2 alone (into pin 7, Discharge) until its voltage falls to (1/3) VCC, at which point the output flips back HIGH and the cycle repeats.

This continuous charge/discharge cycle produces a free-running square wave at the output pin, whose frequency and duty cycle depend only on the values of R1, R2, and C.

Design Equations

Quantity	Formula
Time HIGH (t1)	$t_1 = 0.693 \times (R_1 + R_2) \times C$
Time LOW (t2)	$t_2 = 0.693 \times R_2 \times C$
Total Period (T)	$T = t_1 + t_2 = 0.693 \times (R_1 + 2R_2) \times C$
Frequency (f)	$f = 1.44 / [(R_1 + 2R_2) \times C]$

Duty Cycle	%D = [(R1 + R2) / (R1 + 2R2)] × 100

Calculated Values (R1 = 68 kΩ, R2 = 10 kΩ, C = 10 uF)

Parameter	Calculated Value
Time HIGH (t1)	0.693 × (68k + 10k) × 10uF ≈ 0.541 s
Time LOW (t2)	0.693 × 10k × 10uF ≈ 0.069 s
Total Period (T)	≈ 0.610 s
Frequency (f)	≈ 1.64 Hz (theoretical)
Duty Cycle	≈ 88.6% HIGH

Note: The oscilloscope-measured frequency (see waveform below) is close to 1 Hz, which is slightly lower than the ideal calculated value above. This is expected in practice due to resistor/capacitor tolerances, the electrolytic capacitor's ESR, and breadboard parasitic capacitance — all of which affect the actual RC charge/discharge timing.

Hardware Implementation

The circuit was assembled on a breadboard using the 555 timer IC, the 68 kΩ and 10 kΩ resistors, the 10 uF timing capacitor, and an LED connected to the output pin (through a current-limiting resistor) to visually indicate the square-wave output.

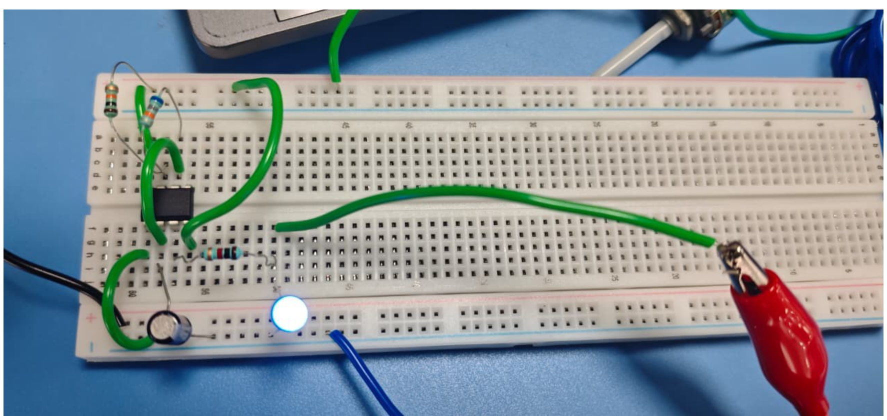

Fig 1: 555 Astable Multivibrator — Breadboard Prototype

Oscilloscope Verification

The output waveform was probed and verified on a Keysight digital storage oscilloscope. The captured waveform confirms a clean, repeating square wave at the output pin, with the measured parameters shown on screen.

Measured Parameter	Oscilloscope Reading
Frequency	974.47 mHz (≈ 0.974 Hz)
Period	1.0262 s

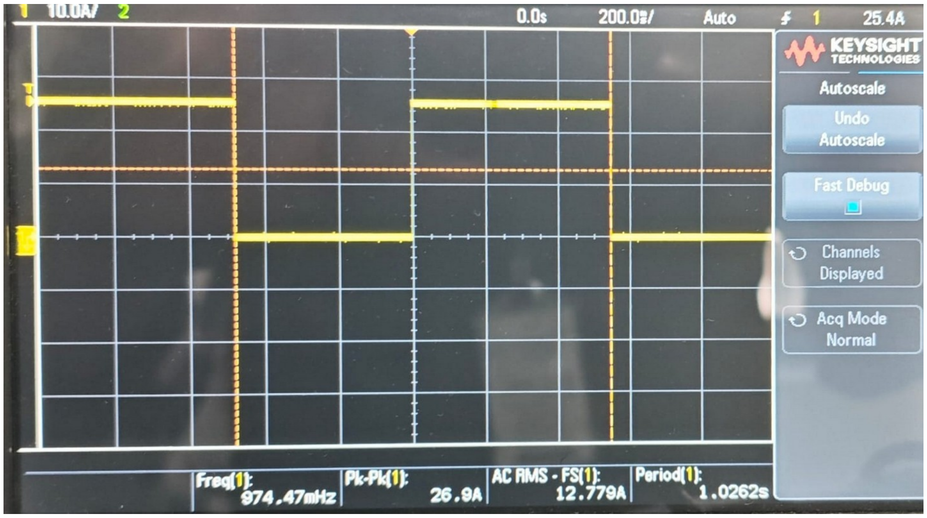

Fig 2: Oscilloscope Capture of the 555 Astable Output Square Wave

Conclusion

The 555 timer astable multivibrator successfully generates a continuous square-wave clock pulse close to 1 Hz, suitable for use as the master time base of the ChronoBit'26 digital clock. This pulse is fed into the CLK input of the Seconds module, from where it propagates through the counter chain to produce the full hours-minutes-seconds and day-of-week timing display.